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June 24, 2024

Details:

$$\dfrac{\dfrac{\dfrac{\dfrac{}{(x \in A) \vdash (x \in A)}\ \text{by}}{(x \in A), (A \subseteq B) \vdash (x \in A)}\ \text{by} \qquad \dfrac{\dfrac{\dfrac{}{(A \subseteq B) \vdash (A \subseteq B)}\ \text{by}}{(A \subseteq B) \vdash (\forall x \in A : x \in B)}\ \text{by}}{(x \in A), (A \subseteq B) \vdash (\forall x \in A : x \in B)}\ \text{by weakening}}{(x \in A), (A \subseteq B) \vdash (x \in B)}\ \text{by}}{\dfrac{(x \in A), (A \subseteq B) \vdash (x \in B)}{(x \in A) \vdash (A \subseteq B) \to (x \in B)}\ \text{by}}\ \text{by}}{\vdash (x \in A) \wedge (A \subseteq B) \to (x \in B)}\ \text{by}$$

Other idea:

$$\dfrac{\dfrac{\dfrac{\dfrac{}{P_3, P_2, P_1 \vdash R}\ \text{Assume/Ident}}{P_3, P_2 \vdash P_1 \to R}\ \text{by}}{P_3 \vdash P_2 \wedge P_1 \to R}\ \text{by}}{\vdash P_3 \wedge P_2 \wedge P_1 \to R}\ \text{by}$$

Details:

$$\dfrac{\dfrac{\dfrac{\dfrac{}{(B \in A) \vdash (B \in A)}\ \text{Assume/Ident}}{(B \in A) \vdash (\exists A : B \in A)}\ \text{by } \exists\text{-intro}}{(B \in A) \vdash (B\ \text{set})}\ \text{by def-intro/set-intro}}{\vdash (B \in A) \to (B\ \text{set})}\ \text{deduction/ImplR/implies-intro}$$